Large Language Model in genetic variant classification

A Focus on Congenital Heart Disease

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Content

Introduction

- Congenital Heart Disease
- Genetic testing workflow
- Challenges in variant interpretation
- ACMG guidelines
- Large Language Model roles

Objectives

- Automating Germline variant interpretation

Congenital Heart Disease data

- Data gathering
- Data preparation

Automating the ACMG Guideline

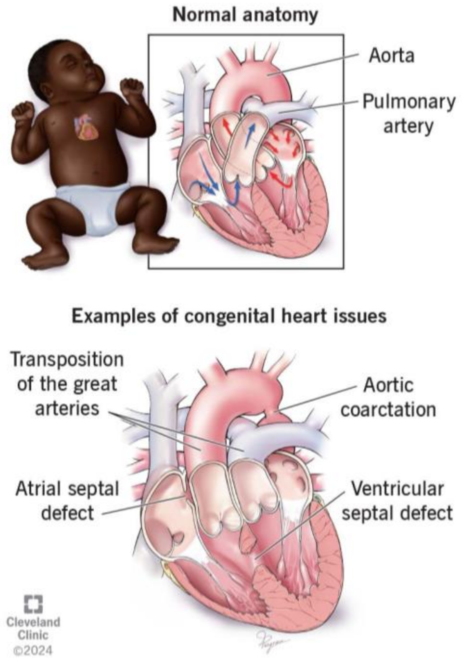
- In-context prompting
- Multi-agent system

Results and Discussion

Future work

Congenital Heart Disease (CHD)

- Congenital heart disease is a general term for a range of **structural problems with the heart** that are present at birth.
- These defects occur when **the heart or its blood vessels do not form correctly** during fetal development.
- They can affect **the normal flow of blood through the heart** and to the rest of the body.
- CHD can range from simple issues, like a small hole in the heart, to complex and life-threatening conditions requiring immediate surgery.
- An estimated **15,000 infants** are born with CHD annually **in Vietnam.**
- Key Challenges: Limited universal newborn screening and a scarcity of pediatric cardiac specialists in rural areas often lead to **delayed diagnosis**.



Morton, S.U. *et al.* Genomic frontiers in congenital heart disease. *Nat Rev Cardiol* **19**, 26–42 (2022).

Congenital Heart Defect NGS Panel for diagnosis

Genes:

ACTC1, ACVR2B, AKT3, ALMS1, ARL6, ARMC4, B9D1, B9D2, BBS1, BBS10, BBS12, BBS2, BBS4, BBS5, BBS7, BBS9, BCOR, BRAF, CBL, CC2D2A, CCDC103, CCDC114, CCDC151, CCDC28B, CCDC39, CCDC40, CCDC65, CCNO, CEP290, CFAP298, CFAP53, CHD7, CITED2, CPS1, CRELD1, CYR61, DNAAF1, DNAAF2, DNAAF3, DNAAF4, DNAAF5, DNAH11, DNAH5, DNAH8, DNAI1, DNAI2, DNAL1, DRC1, DTNA, ELN, FLNA, FOXF1, FOXH1, GATA4, GATA6, GDF1, GJA1, GPC3, HAND1, HRAS, INVS, JAG1, KIF7, KRAS, LEFTY2, MAP2K1, MAP2K2, MCIDAS, MED13L, MKKS, MKS1, MMP21, MYH6, NEK8, NKX2-5, NKX2-6, NME8, NODAL, NOTCH1, NPHP3, NR2F2, NRAS, NSD1, NTRK3, OFD1, PIK3CA, PIK3R2, PITX2, PKD1L1, PTPN11, RAF1, RAI1, RBM10, RIT1, RPGRIP1L, SEMA3E, SHOC2, SMAD6, SOS1, SPAG1, TAB2, TBX1, TBX20, TBX5, TCTN2, TLL1, TMEM216, TMEM231, TMEM67, TRIM32, TTC8, WDPCP, ZFPM2, ZIC3, ZMYND10 (115 genes)

Coverage:

≥99% at 20x

Turnaround Time:

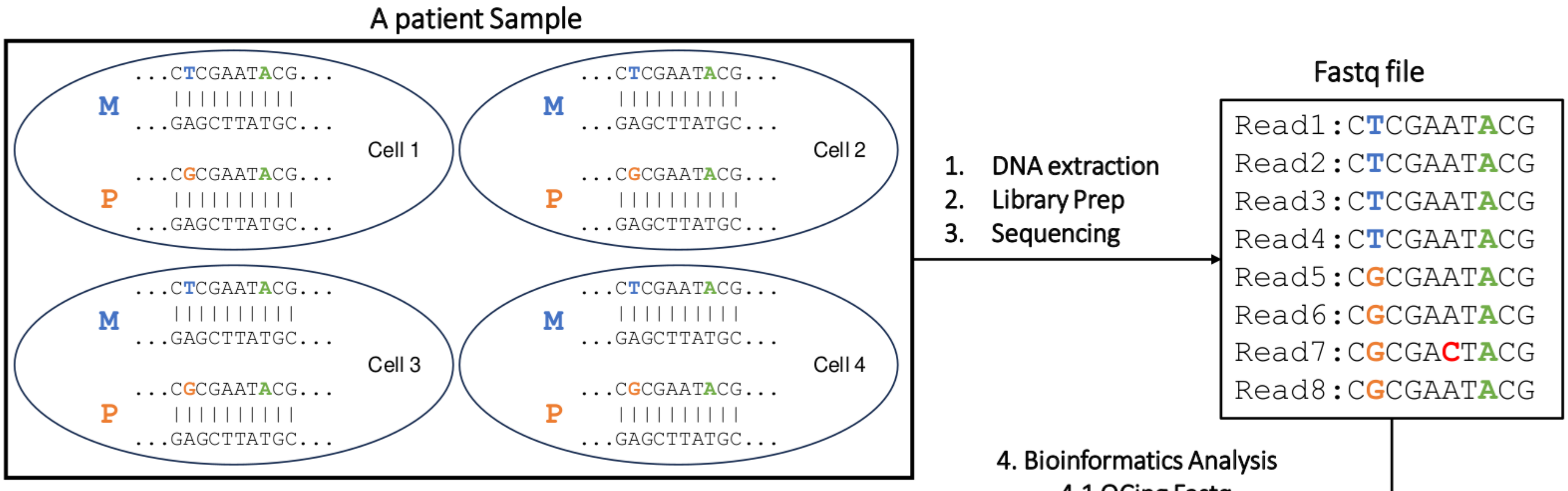
2 - 3 weeks

Specimen Requirements:

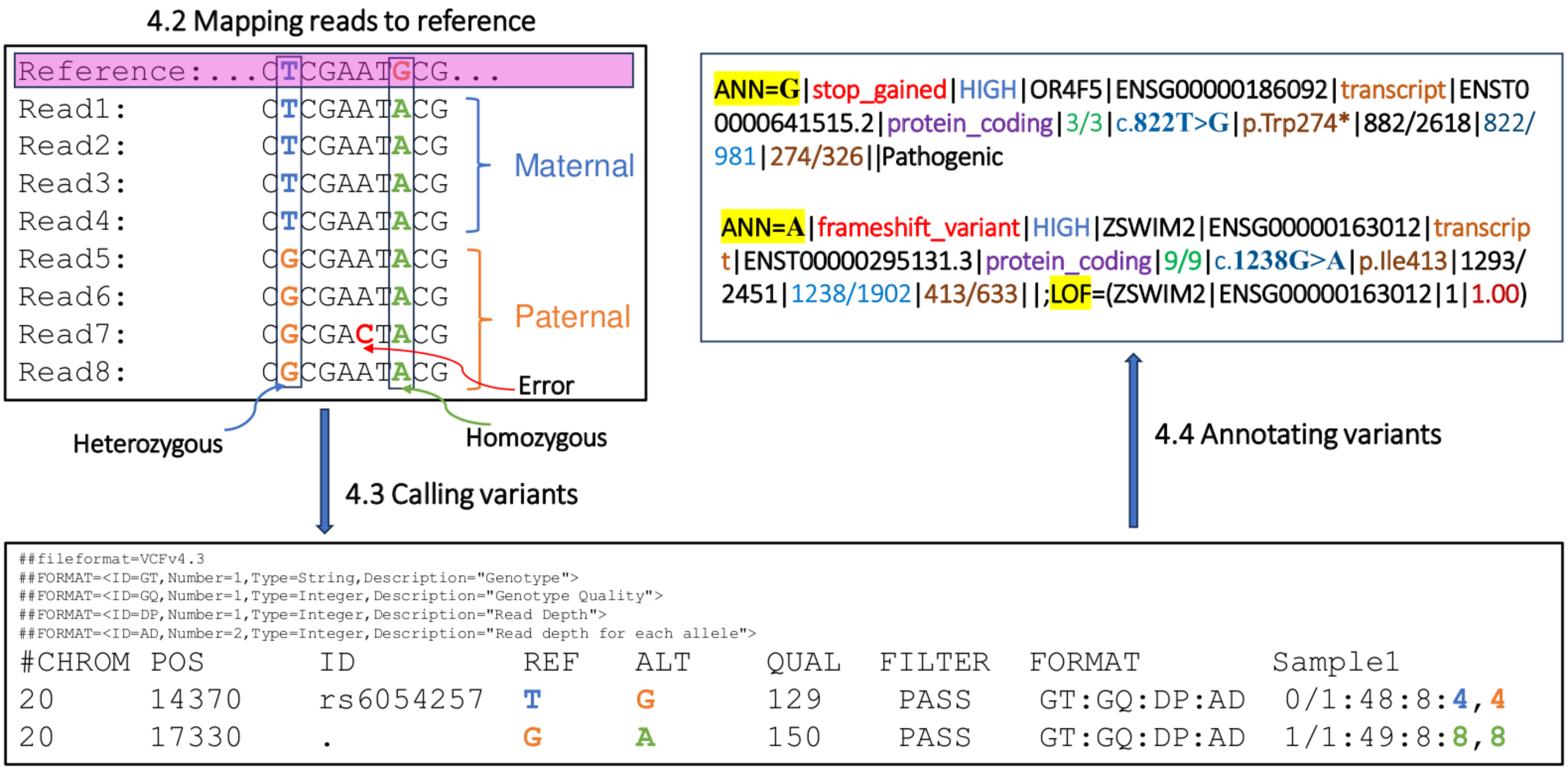
Blood (two 4ml EDTA tubes, lavender top) or Extracted DNA (3ug in EB buffer) or Buccal Swab or Saliva (kits available upon request)

https://www.fulgentgenetics.com/congenital-heart-defect

Congenital Heart Defect NGS Panel: steps



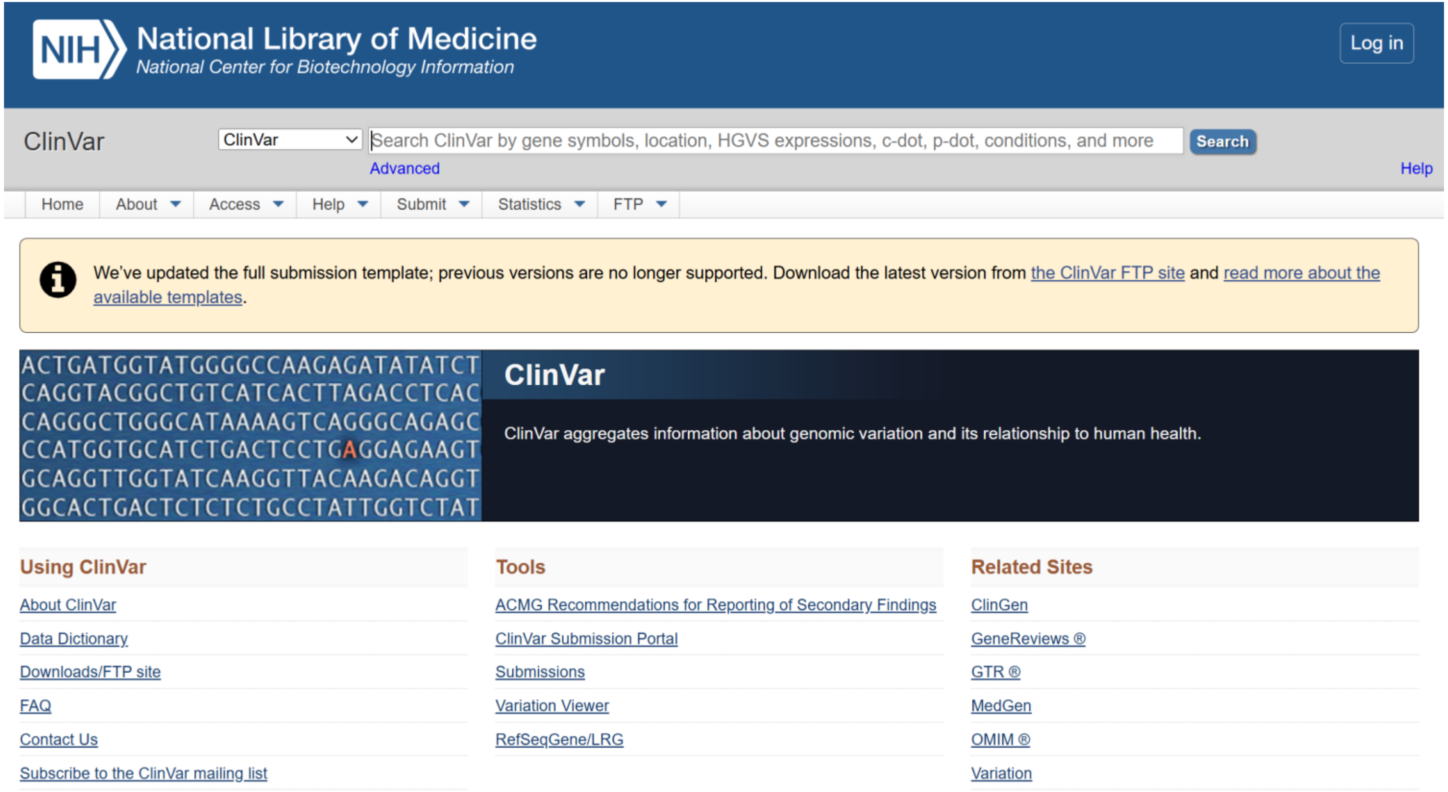
Congenital Heart Defect NGS Panel: steps



Genome Board: interpretation of variants

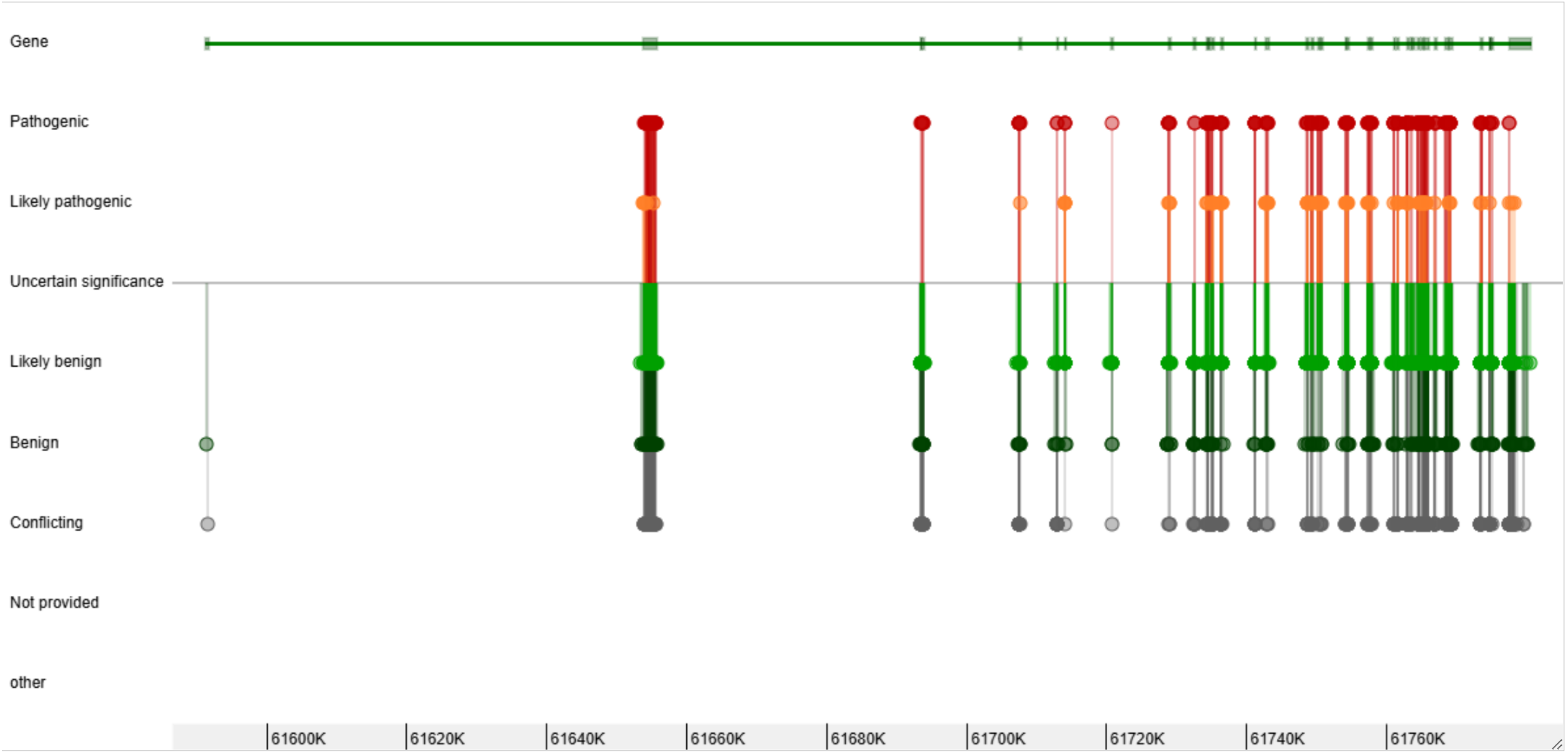
https://ggba.swiss/en/the-first-medical-genomics-center-opens-in-geneva/

ClinVar: Human variations classified for diseases and drug responses

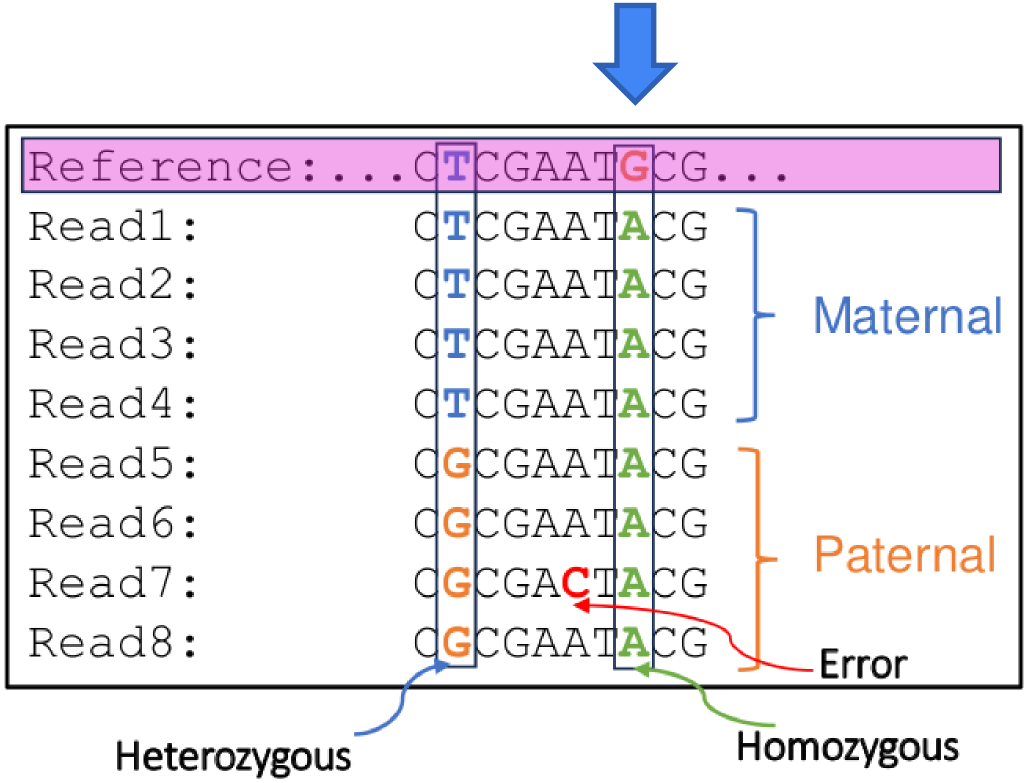


ClinVar: Human variations classified for diseases and drug responses

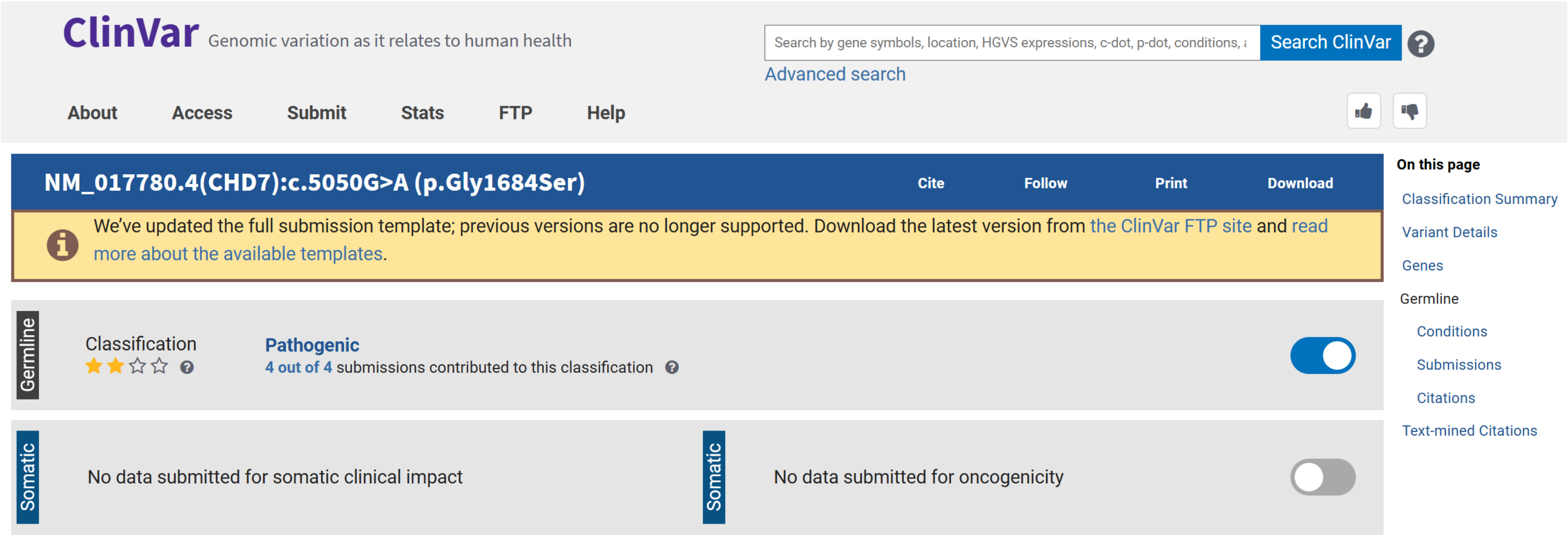
CHD7



Annotation of CHD7 variants using ClinVar



Annotation of CHD7 variants using ClinVar



Annotation of CHD7 variants using ClinVar

Identifiers: NM_017780.4(CHD7):c.5050G>A (p.Gly1684Ser)

Variation ID: 520773 Accession: VCV000520773.15

Type and length: single nucleotide variant, 1 bp

Location: Cytogenetic: 8q12.2 8: 60845063 (GRCh38) [NCBI UCSC] 8: 61757622 (GRCh37) [NCBI UCSC]

Timeline in ClinVar:

	First in ClinVar	Last submission	Last evaluated
Germline	Apr 15, 2018	Apr 13, 2025	Sep 10, 2024

HGVS:

Nucleotide	Protein	Molecular consequence
NM_017780.4:c.5050G>A MANE SELECT	NP_060250.2:p.Gly1684Ser	missense
NM_001316690.1:c.1717-17166G>A		intron variant
NC_000008.11:g.60845063G>A		

... more HGVS

Protein change: G1684S

Other names: -

Canonical SPDI: NC_000008.11:60845062:G:A

Global minor allele frequency (GMAF): -

Allele frequency: -

Links:
ClinGen: CA371320188
dbSNP: rs1554602465
VarSome

Annotation of CHD7 variants using ClinVar

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Expand all rows
Pathogenic (Sep 25, 2015) C Contributing to aggregate classification	★☆☆☆ (Ambry exome assertion method (8-5-2015))	Inborn genetic diseases	Ambry Genetics Accession: SCV000741041.4 First in ClinVar: Apr 15, 2018 Last updated: Apr 13, 2025	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: yes • more
Pathogenic (Sep 10, 2024) C Contributing to aggregate classification	★☆☆☆ (Invitae Variant Classification Sherloc (09022015))	CHARGE syndrome	Labcorp Genetics (formerly Invitae), Labcorp Accession: SCV001580653.5 First in ClinVar: May 10, 2021 Last updated: Feb 25, 2025	Publications: PubMed (6) Comment: show Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: unknown
Pathogenic (Dec 11, 2021) C Contributing to aggregate classification	★☆☆☆ (GeneDx Variant Classification Process June 2021)	Not Provided	GeneDx Accession: SCV001794204.3 First in ClinVar: Aug 21, 2021 Last updated: Mar 04, 2023	Comment: show Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: yes
Pathogenic (Jul 13, 2022) C Contributing to aggregate classification	★☆☆☆ (ACMG Guidelines, 2015)	not provided	Clinical Genetics Laboratory, Skane University Hospital Lund Accession: SCV005197484.1 First in ClinVar: Aug 25, 2024 Last updated: Aug 25, 2024	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: yes

Annotation of CHD7 variants using ClinVar

Citations for germline classification of this variant

Title	Author	Journal	Year	Link
Variations in Multiple Syndromic Deafness Genes Mimic Non-syndromic Hearing Loss.	Bademci G *et al.*	Scientific reports	2016	PMID: 27562378
The prevalence of CHD7 missense versus truncating mutations is higher in patients with Kallmann syndrome than in typical CHARGE patients.	Marcos S *et al.*	The Journal of clinical endocrinology and metabolism	2014	PMID: 25077900
Phenotype in 18 Danish subjects with genetically verified CHARGE syndrome.	Husu E *et al.*	Clinical genetics	2013	PMID: 22462537
Mutation update on the CHD7 gene involved in CHARGE syndrome.	Janssen N *et al.*	Human mutation	2012	PMID: 22461308
Aberrant 5' splice sites in human disease genes: mutation pattern, nucleotide structure and comparison of computational tools that predict their utilization.	Buratti E *et al.*	Nucleic acids research	2007	PMID: 17576681
Statistical features of human exons and their flanking regions.	Zhang MQ	Human molecular genetics	1998	PMID: 9536098

ACMG guidelines

	Benign		Pathogenic			
	Strong	Supporting	Supporting	Moderate	Strong	Very strong
Population data	MAF is too high for disorder BA1/BS1 **OR** observation in controls inconsistent with disease penetrance BS2			Absent in population databases PM2	Prevalence in affecteds statistically increased over controls PS4	
Computational and predictive data		Multiple lines of computational evidence suggest no impact on gene /gene product BP4 Missense in gene where only truncating cause disease BP1 Silent variant with non predicted splice impact BP7 In-frame indels in repeat w/out known function BP3	Multiple lines of computational evidence support a deleterious effect on the gene /gene product PP3	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before PM5 Protein length changing variant PM4	Same amino acid change as an established pathogenic variant PS1	Predicted null variant in a gene where LOF is a known mechanism of disease PVS1
Functional data	Well-established functional studies show no deleterious effect BS3		Missense in gene with low rate of benign missense variants and path. missenses common PP2	Mutational hot spot or well-studied functional domain without benign variation PM1	Well-established functional studies show a deleterious effect PS3	
Segregation data	Nonsegregation with disease BS4		Cosegregation with disease in multiple affected family members PP1	Increased segregation data		
De novo data				De novo (without paternity & maternity confirmed) PM6	De novo (paternity and maternity confirmed) PS2	
Allelic data		Observed in *trans* with a dominant variant BP2 Observed in *cis* with a pathogenic variant BP2		For recessive disorders, detected in trans with a pathogenic variant PM3		
Other database		Reputable source w/out shared data = benign BP6	Reputable source = pathogenic PP5			
Other data		Found in case with an alternate cause BP5	Patient's phenotype or FH highly specific for gene PP4			

There are **28 criteria** in the ACMG guidelines. During variant interpretation, variants are classified into **five tiers: Pathogenic (P), Likely pathogenic (LP), Uncertain significance (VUS), Likely benign (LB), and Benign (B),** depending on the applied criteria. 28 criteria can be classified by the weight and type of evidence indicated by each criterion.

- Classification by **evidence weight**

  The 28 criteria can be classified into two: **16 Pathogenic criteria and 12 Benign criteria.** Depending on the weight of evidence, the pathogenic criteria are divided into very strong (PVS1), strong (PS1–4), moderate (PM1–6), supporting (PP1–5); the benign criteria are divided into stand-alone (BA1), strong (BS1–4), supporting (BP1–7). Criteria within the same category are given the same weight.

- Classification by **evidence type**

  The 28 criteria can be classified into **8 types: population data, computational data, functional data, segregation data, *de novo* data, allelic data, other database, and other data,** depending on the source of evidence. You can check the details of each criteria in Figure 1 and the ACMG guidelines.

Ref: https://3billion.io/blog/what-are-the-acmg-standards-and-guidelines-and-how-do-they-work

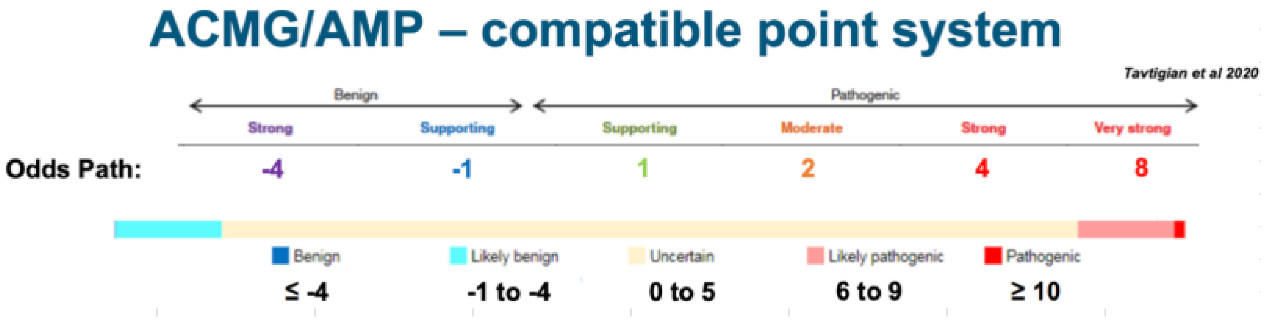
Scoring and Classification

ACMG Scoring System

- Very Strong (PVS): +8 points
- Strong (PS): +4 points
- Moderate (PM): +2 points
- Supporting (PP): +1 point
- Stand-alone Benign (BA): -8 points
- Strong Benign (BS): -4 points
- Supporting Benign (BP): -1 point

Classification Based on Total Score

- Pathogenic: Total score ≥10.
- Likely Pathogenic: Total score 6-9.
- Uncertain Significance (VUS): Total score 0-5.
- Likely Benign: Total score -1 to -6.
- Benign: Total score ≤ -7.



The Challenge of Variant Interpretation following the ACMG-AMP 2015

- **Complexity:** Interpreting germline variants is a major challenge in clinical genetics.
- **Guidelines:** The process follows the ACMG-AMP 2015 and ACGS 2020 best practice guidelines.
- **Current Limitations:** Existing annotation tools are incomplete or insufficient.
- **Need for Automation:** Better tools are needed to provide clear, consistent, and reproducible variant classifications.
- **Focus:** This project applies automated interpretation methods to Congenital Heart Disease (CHD).

Why Large Language Models (LLM)?

1. Solving tasks **without explicit programming** (checking 28 criteria via prompting).

2. Understanding **large volumes of text** that would take longer to read manually (review PubMed articles).

3. Handling new scenarios without predefined rules (AI Agents are able to break down goals into actionable steps).

CHD data profile (189 genes, 408,231 variants)

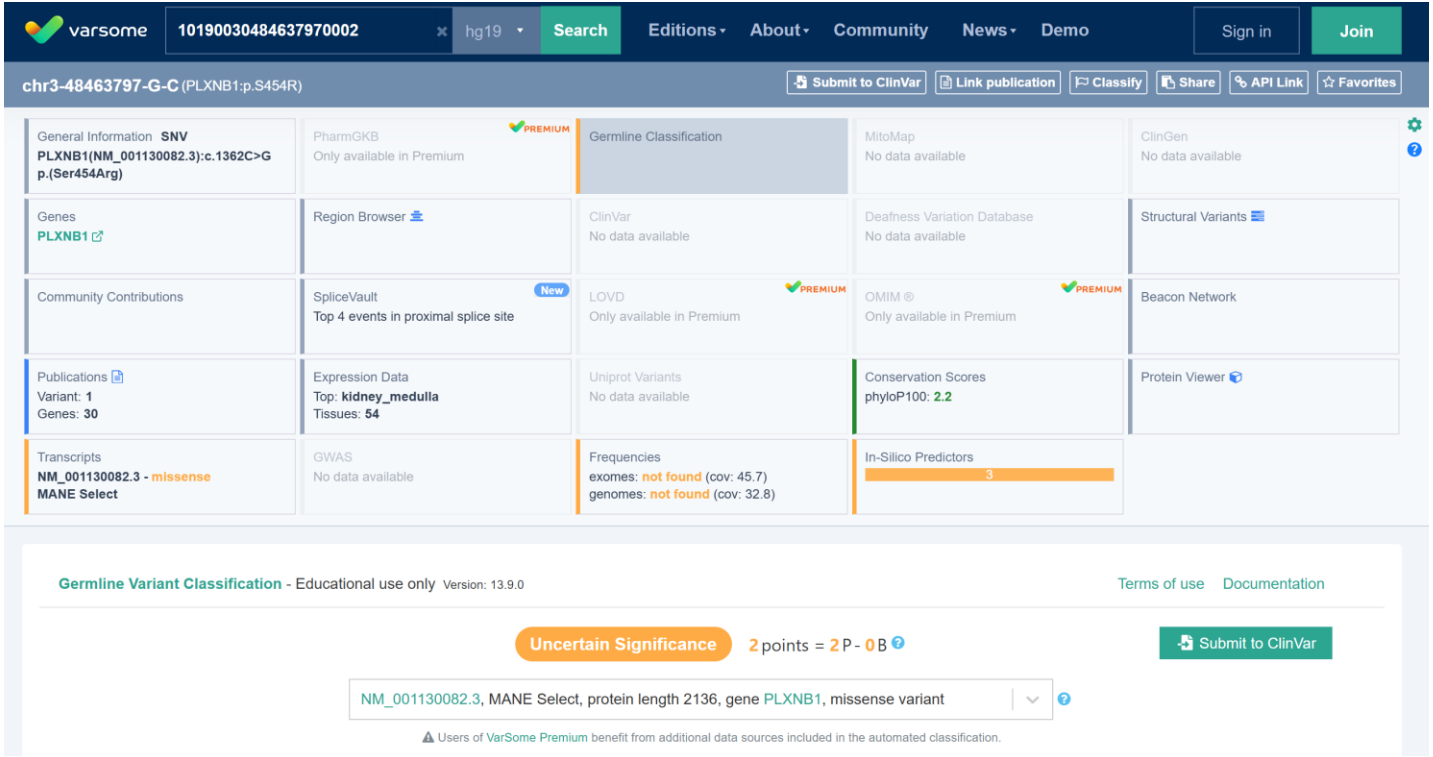
Definitions | Toggle filter | Reset filter | Download table

Gene	CHD classification	Extra cardiac phenotype	Inheritance mode	Ranking	Supporting References
BCOR	ASD with minor abnormalities; VSD with minor abnormalities; Obstructive lesions; Malformation of outflow tracts; Heterotaxy	✓	XLD	★★★★★	2
BMPR2	ASD with minor abnormalities; VSD with minor abnormalities; AVSD and variants; Malformation of outflow tracts	?	AD DN	★★★★★	3
BRAF	ASD with minor abnormalities; Obstructive lesions	✓	AD	★★★★★	5
CDK13	ASD; VSD; Obstructive lesions	✓	AD	★★★★★	2
CFC1	AVSD and variants; Obstructive lesions; Malformation of outflow tracts; Heterotaxy	✓	AD	★★★★★	4
CHD4	ASD; VSD; Obstructive lesions; Malformation of outflow tracts	✓	AD	★★★★★	2
CHD7	ASD with minor abnormalities	✓	AD	★★★★★	3
CHST14	ASD; Obstructive lesions	✓	AR	★★★★★	3
CITED2	ASD; VSD	?	AD	★★★★★	3
CREBBP	ASD with minor abnormalities; VSD with minor abnormalities; Obstructive lesions	✓	AD	★★★★★	6

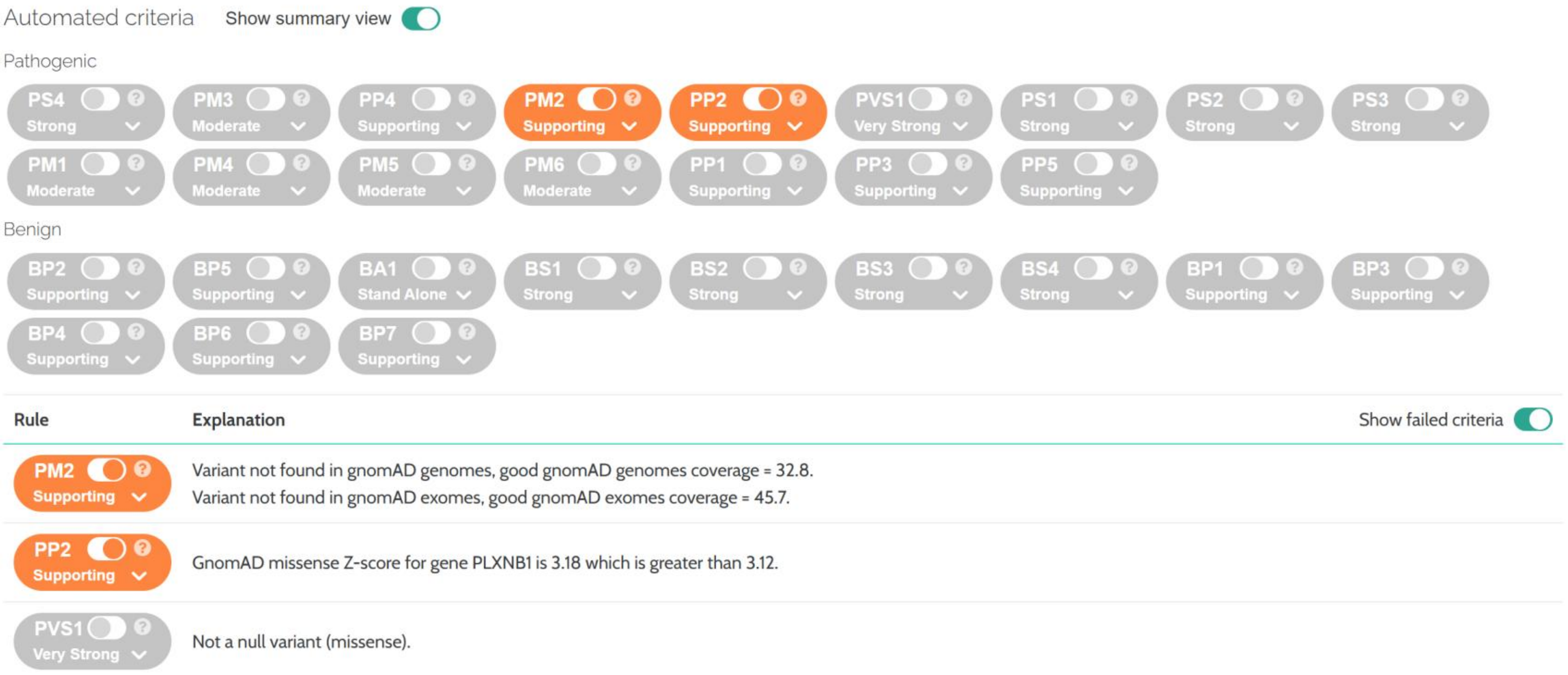
11 - 20 / 189 (189) Jump to page: 2 Genes per page: 10

https://chdgene.victorchang.edu.au/
https://www.ahajournals.org/doi/10.1161/CIRCGEN.121.003539

Varsome Variant Interpretation as gold standard



Varsome Variant Interpretation as gold standard

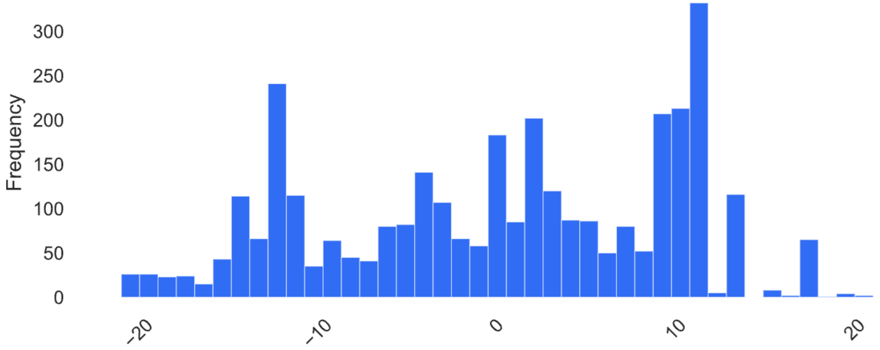


Data Preparation CHD (170 genes, 3312 variants)

ACMG class

Value	Count	Frequency (%)
Benign	878	26.5%
Uncertain Significance	763	23.0%
Pathogenic	748	22.6%
Likely Benign	534	16.1%
Likely Pathogenic	389	11.7%

ACMG score



Histogram with fixed size bins (bins=41)

ACMG criteria

	name	count
0	PM2	2317
1	PP5	1177
2	BP4	1165
3	BP6	1139
4	PVS1	750
5	BP7	663
6	PM1	639
7	BS2	585
8	BP1	396
9	BA1	394
10	PM5	380
11	BS1	323
12	PP2	170
13	PP3	71
14	PS1	40
15	BP3	37
16	PS3	25
17	BS3	2

```
Gene not included: 19

{'ADAMTS19',
 'AMOTL1',
 'B3GAT3',
 'CCDC22',
 'CIROP',
 'CITED2',
 'HAND2',
 'MAPKAPK5',
 'ODAD4',
 'PAN2',
 'PRDM6',
 'PRKACA',
 'PRKACB',
 'RBFOX2',
 'SMG9',
 'SPRED2',
 'UBR7',
 'ZIC3',
 'ZNF699'}
```

Evaluation Metrics

- **Pearson correlation:** Measures the correlation between true and predicted ACMG/ACGS scores.
- **Confusion Matrix:** Visualizes the accuracy of classification by comparing true and predicted labels across categories (Benign, Likely Benign, VUS, Likely Pathogenic, Pathogenic).
- **Jaccard Similarity:** Quantifies the overlap between the true and predicted sets of ACMG/ACGS criteria applied. A score of 0.600 indicates a 60% overlap.

Jaccard Similarity

$|A \cap B| / |A \cup B|$

Example:

True:{'PVS1', 'PM2', 'PP5'}

Predicted:{'PVS1', 'PM2', 'PP3', 'BP4', 'BP7'}

Jaccard similarity = **0.33**

#1 Compute the intersection

Intersection = True ∩ Predicted
= { PVS1, PM2 } = 2

#2 Compute the union

Union = True ∪ Predicted
= { PVS1, PM2, PP5, PP3, BP4, BP7 }
= 6

#3 Plug into the Jaccard formula

Jaccard = 2/6 = 0.33

#4 Interpretation

A Jaccard score of 0.33 means 33% overlap between the true and predicted criterion sets.

Baseline (deepseek-chat)

deepseek

🧬 TMEM94
c.765-1G>C

```
Analyze this variant: `{variant_name}` and list all met ACMG criteria with their strengths in a JSON format:

Provide the response in this exact JSON format:
{{
    "classifications": [
        {{
            "name": "<criteria_name>",
            "met_criteria": true,
            "user_explain": [
                "<explanation1>",
                "<explanation2>"
            ],
            "strength": "<strength_level>"
        }}
    ]
}}

Only use criteria from this list: PM2, PP5, BP4, BP6, PVS1, BP7, PM1, BS2, BA1, BP1, PM5, BS1, PP2, PP3, PS1,
                                  BP3, PS3, BS3
Only use strength levels from: Supporting, Moderate, Strong, Very strong
```

In-context Prompting 🤖

Final Classification & Explanation

Baseline: in-context prompting

```
Analyze this variant: `{variant_name}` and list all met ACMG criteria with their strengths in a JSON format:

Provide the response in this exact JSON format:
{{
    "classifications": [
        {{
            "name": "<criteria_name>",
            "met_criteria": true,
            "user_explain": [
                "<explanation1>",
                "<explanation2>"
            ],
            "strength": "<strength_level>"
        }}
    ]
}}

Only use criteria from this list: PM2, PP5, BP4, BP6, PVS1, BP7, PM1, BS2, BA1, BP1, PM5, BS1, PP2, PP3, PS1,
                                  BP3, PS3, BS3
Only use strength levels from: Supporting, Moderate, Strong, Very strong
```

We initially aim to work on 18 criteria

Baseline: in-context prompting (example)

```
Analyze this variant: `NM_014738.6(TMEM94):c.765-1G>C` and list all met ACMG criteria with their strengths
in a JSON format:

Provide the response in this exact JSON format:
{{
    "classifications": [
        {{
            "name": "<criteria_name>",
            "met_criteria": true,
            "user_explain": [
                "<explanation1>",
                "<explanation2>"
            ],
            "strength": "<strength_level>"
        }}
    ]
}}

Only use criteria from this list: PM2, PP5, BP4, BP6, PVS1, BP7, PM1, BS2, BA1, BP1, PM5, BS1, PP2, PP3, PS1,
                                  BP3, PS3, BS3
Only use strength levels from: Supporting, Moderate, Strong, Very strong
```

Baseline: in-context prompting (example)

```
{
  "variant_name": "NM_014738.6(TMEM94):c.765-1G>C",
  "classifications": [
    {
      "name": "PVS1",
      "met_criteria": true,
      "user_explain": [
        "The variant affects the canonical splice site at position -1 of the intron, which is highly likely to disrupt
        splicing.",
        "Splice site variants at -1 positions are predicted to have a significant impact on gene function."
      ],
      "strength": "Very strong"
    },
    {
      "name": "PM2",
      "met_criteria": true,
      "user_explain": [
        "The variant is absent from population databases (e.g., gnomAD), indicating it is rare in the general population."
      ],
      "strength": "Moderate"
    },
```

Hallucinated!

```
    {
      "name": "PP3",
      "met_criteria": true,
      "user_explain": [
        "Multiple computational predictors (e.g., SpliceAI, MaxEntScan) support a deleterious effect on splicing."
      ],
      "strength": "Supporting"
    },
```

Hallucinated!

```
    {
      "name": "BP4",
      "met_criteria": false,
      "user_explain": [
        "No computational evidence suggests a benign effect."
      ],
      "strength": "N/A"
    },
    {
      "name": "BP7",
      "met_criteria": false,
      "user_explain": [
        "The variant is not a silent intronic change outside of the splice site consensus sequence."
      ],
      "strength": "N/A"
    }
  ]
}
```

LLM prediction
Score = 8 + 2 + 1 + 0 + 0 = 11
Pathogenic: Total score ≥10.

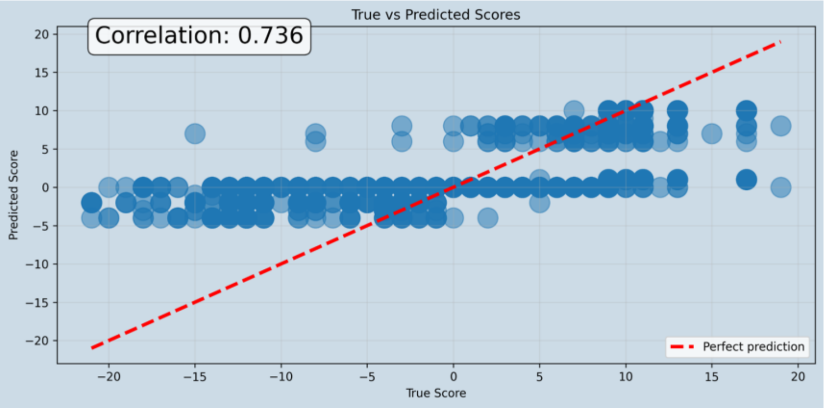
Pathogenic 10 points = 10 P - 0 B

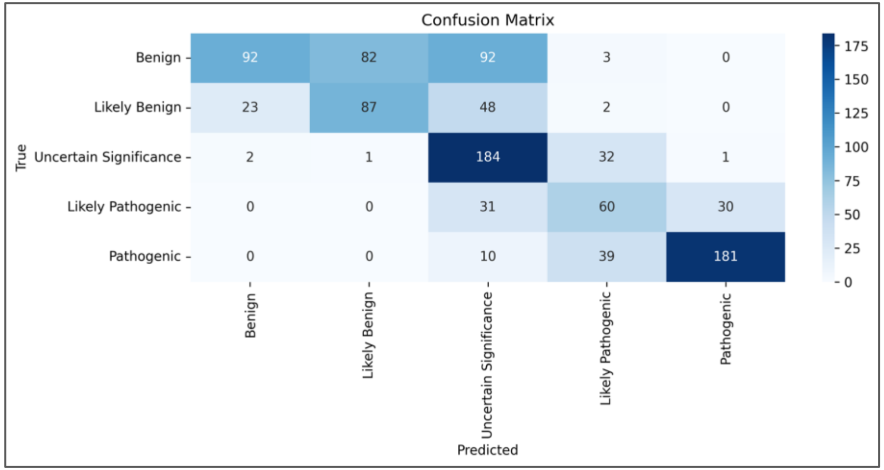
Rule	Explanation
PVS1 Very Strong	Null variant (intronic within ±2 of splice site) in gene TMEM94. Loss-of-function is a known mechanism of disease (gene has 22 reported pathogenic LOF variants).
PM2 Supporting	Variant not found in gnomAD genomes, good gnomAD genomes coverage = 33.5. Variant not found in gnomAD exomes, good gnomAD exomes coverage = 64.6.
PP5 Supporting	Supporting: ClinVar classifies this variant as Pathogenic, 1 star (reviewed Apr '25, 2 submissions), citing 30526868.

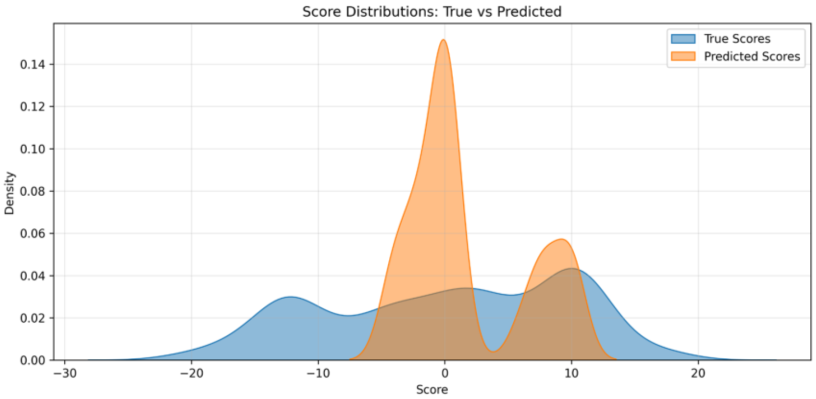
Show failed criteria

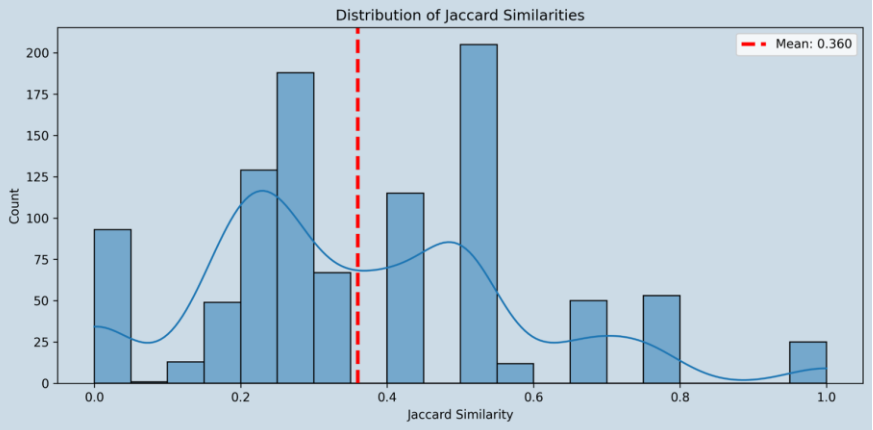
Varsome: https://varsome.com/variant/hg19/rs1352010373?

Baseline (deepseek-chat)









Hallucination example

Can you help me analyze this variant: NM_138927.4(SON):c.3662C>T (p.Ser1221Leu)

Output
A validation report written for human readable with the following points:
- **Gerline classification**:
  - List of met criteria: name, strength, explanation in table format
  - Total ACMG score
  - Final classification: The validated ACMG classification (e.g., Likely Benign).

Of course. Here is a detailed validation report for the variant **NM_138927.4(SON):c.3662C>T (p.Ser1221Leu)**.

Variant Validation Report

Variant: NM_138927.4(SON):c.3662C>T (p.Ser1221Leu)
Gene: SON
HGVS Protein Change: p.Ser1221Leu

Germline Classification

The variant is classified as a **Variant of Uncertain Significance (VUS)**.

Total ACMG Score: 0 (No points for or against pathogenicity)

List of Met Criteria

Criterion Code	Strength	Direction	Explanation
PM2_Supporting	Supporting	Pathogenic	The variant is absent from population databases (gnomAD v2.1.1 and v4.0.0), indicating it is not a common benign polymorphism.
BP1	Supporting	Benign	The SON gene is not associated with a dominant Mendelian disorder with a known mechanism of disease. Missense variants are not a common cause of disease in this gene.
BP4	Supporting	Benign	Multiple lines of computational evidence (in silico prediction tools) suggest this missense variant has no impact on the protein.

LLM limitations

1. Limited access to real-world data: **knowledge is restricted** to the training cut-off date (e.g. cannot update new data from ClinVar, gnomAD).
2. **Hallucinations**: generate plausible but false information (e.g. predict plausible next words but incorrect)
3. **Context window** constraints: limited memory for long conversations or large documents.
4. Struggles with **complex reasoning** – May fail at multi-step logical or mathematical tasks.

Our approaches to resolve the LLM limitations

Problem	Solution
1. Limited access to real-world data	**1. Knowledge Integration via APIs** • Tool-augmented accuracy: Fetch real-time data from ClinVar, gnomAD, UniProt to ground interpretations in evidence. • Dynamic updates: Overcome training cutoffs by querying latest research/classifications. • Standardized evidence: Map API outputs to ACMG/ACGS criteria (e.g., PS1, PM2).
2. Hallucinations	**2. Evidence-Driven Automation** • Rule-based reasoning: Automate ACMG classification steps (e.g., PVS1 for loss-of-function). • Uncertainty flags: Highlight low-confidence calls needing manual review. • Audit trails: Log evidence sources for reproducibility (e.g., "Pathogenic per ClinVar X/Y submissions").

Our approaches to resolve the LLM limitations

Problem	Solution
3. Context window constraints	**3. Optimized LLM Workflow** • Prompt engineering: Templates for consistent guideline adherence (e.g., "Extract POPMAX from gnomAD"). • Memory management: Cache frequent queries (e.g., common variants) to reduce API costs/latency. • Hybrid logic: Combine LLM flexibility with deterministic rules (e.g., "If allele frequency > 5%, auto-classify as BENIGN").
4. Struggles with complex reasoning	**4. Multi-Agent Coordination** • Specialist agents: Divide tasks (e.g., one agent handles PubMed searches, another computes ACMG scores). • Conflict resolution: Cross-check disagreements (e.g., ClinVar vs. UniProt). • Pipeline design: Orchestrate workflows (e.g., "Variant → Annotation → Classification → Report").

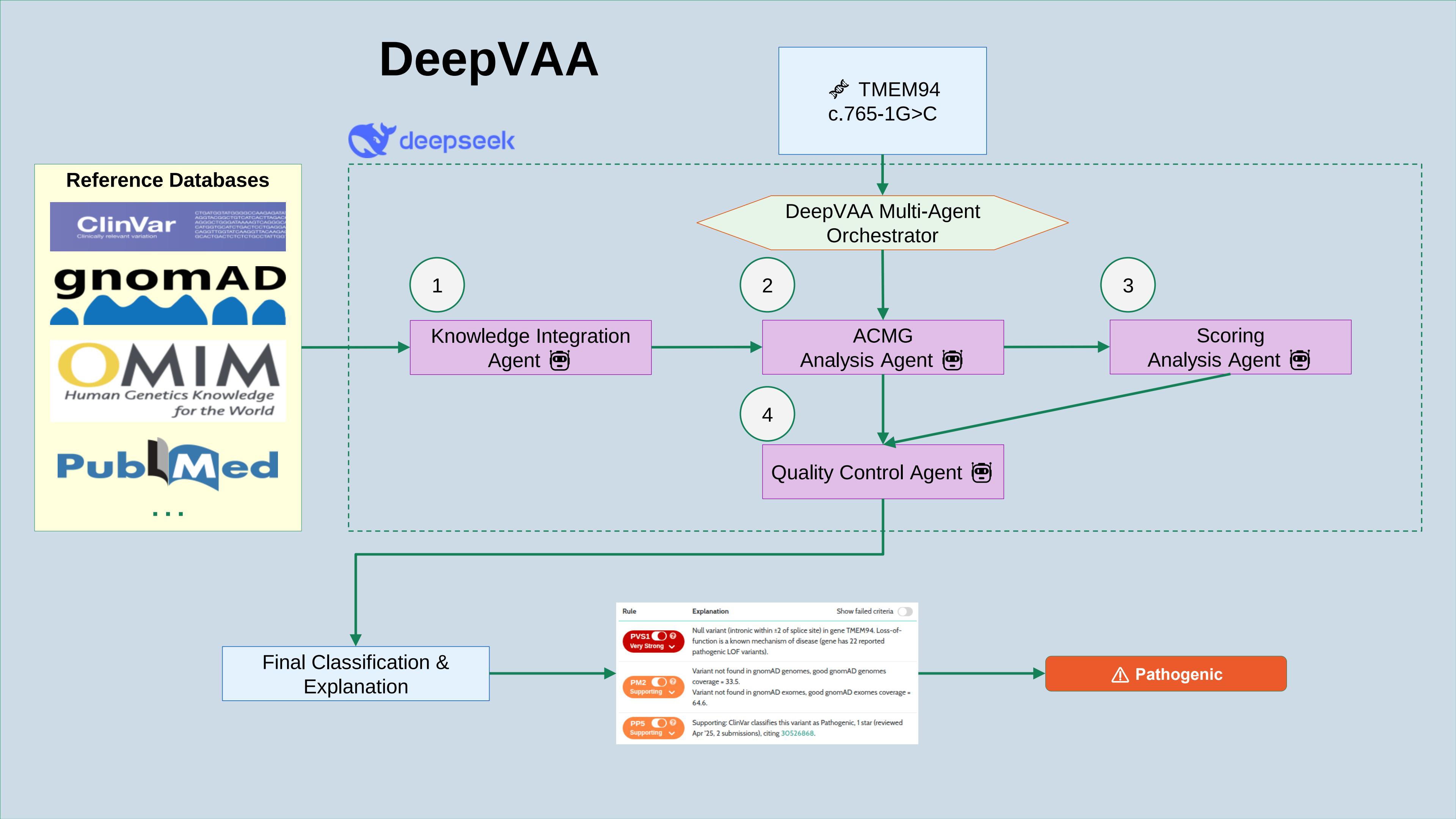
DeepVAA System Architecture: A Multi-Agent Approach

DeepVAA (Deepseek Variant Automation and Annotation) System: A novel multi-agent system designed to automate and interpret genetic variants.

Key Objective: To provide comprehensive variant annotation and interpretation through orchestrated specialized agents.

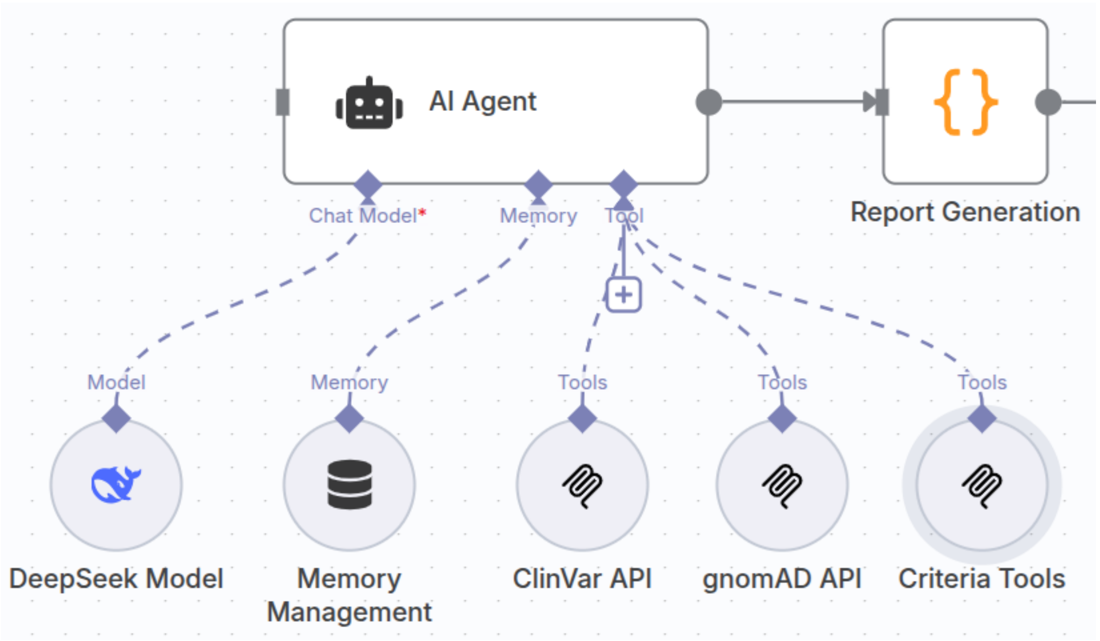
Four Specialized Agents:

- Knowledge Integration Analyst (Knowledge Agent): Synthesizes structured variant data and consolidates evidence from various sources. This agent's goal is to extract, consolidate, and summarize evidence from ClinVar, gnomAD, and other external sources.
- ACMG Analysis Agent (ACMG Agent): Responsible for evaluating all 28 ACMG-AMP 2015 criteria based on available clinical, population, computational, and functional evidence.
- ACGS Scoring Specialist (ACGS Agent): Independently analyzes a variant's ACMG criteria to compute its point-based classification according to the ACGS 2020 Best Practice Guidelines.
- Quality Control Auditor (Quality Control Agent): Serves as a quality control gatekeeper, ensuring annotations are consistent, justified, and reproducible.



DeepVAA
TMEM94
c.765-1G>C
deepseek
Reference Databases
ClinVar
Clinically relevant variation
gnomAD
OMIM
Human Genetics Knowledge for the World
PubMed
...
DeepVAA Multi-Agent Orchestrator
1
2
3
4
Knowledge Integration Agent
ACMG Analysis Agent
Scoring Analysis Agent
Quality Control Agent
Final Classification & Explanation
Rule
Explanation
Show failed criteria
PVS1 Very Strong
Null variant (intronic within ±2 of splice site) in gene TMEM94. Loss-of-function is a known mechanism of disease (gene has 22 reported pathogenic LOF variants).
PM2 Supporting
Variant not found in gnomAD genomes, good gnomAD genomes coverage = 33.5.
Variant not found in gnomAD exomes, good gnomAD exomes coverage = 64.6.
PP5 Supporting
Supporting: ClinVar classifies this variant as Pathogenic, 1 star (reviewed Apr '25, 2 submissions), citing 30526868.
Pathogenic

Architecture of an AI Agent



Model: DeepSeek Chat Model used for agent reasoning.

Memory: a storage system for retaining information.

Tool: APIs calling: "ClinVar API" and "gnomAD API," are external tools the AI Agent can access external world data.

1. Knowledge Agent

Preview

Formatted JSON

Input

system: You are Knowledge Integration Analyst. You synthesize structured variant data into consolidated evidence summaries for variant interpretation.
Your personal goal is: Extract, consolidate, and summarize evidence from ClinVar, gnomAD, and external sources
You ONLY have access to the following tools, and should NEVER make up tools that are not listed here:

Tool Name: clinvar_variant_lookup
Tool Arguments: {'gene': {'description': 'Gene symbol, e.g. BRCA1', 'type': 'str'}, 'hgvs': {'description': 'HGVS c. notation, e.g. c.68_69delAG', 'type': 'str'}}
Tool Description: Searches ClinVar using gene and HGVS c. notation, returns clinical interpretation summary.
Tool Name: gnomad_variant_summary
Tool Arguments: {'gene': {'description': 'Gene symbol, e.g. BRCA1', 'type': 'str'}, 'hgvsc': {'description': 'HGVSc variant filter, e.g. c.4357+1G>A', 'type': 'Union[str, NoneType]'}}
Tool Description: Query gnomAD for gene-level variant data and filter by HGVSc if needed

IMPORTANT: Use the following format in your response:

```
Thought: you should always think about what to do
Action: the action to take, only one name of [clinvar_variant_lookup, gnomad_va
Action Input: the input to the action, just a simple JSON object, enclosed in c
Observation: the result of the action
```

Once all necessary information is gathered, return the following format:

```
Thought: I now know the final answer
Final Answer: the final answer to the original input question
```

user:
Current Task:
Consolidate all available knowledge about the variant SNV ACAD9(NM_014049.5):c.976G>A p.(Ala326Thr).
Include ClinVar classification, gnomAD frequencies, consequence type, and any known RSIDs.
Provide a structured evidence summary suitable for downstream ACMG and ACGS interpretation.

This is the expected criteria for your final answer: Consolidated evidence summary in bullet-point format.

Output

Thought: I now know the final answer
Final Answer:

- **Variant**: ACAD9 (NM_014049.5):c.976G>A p.(Ala326Thr)
- **ClinVar Interpretation**:
  - Clinical Significance: Benign/Likely benign
  - Review Status: Criteria provided, multiple submitters, no conflicts
  - Last Evaluated: 2025-03-11
  - ClinVar Accession: VCV000136253.35
  - RCV Accessions: RCV000123467, RCV000395019, RCV000676686
- **gnomAD Data**:
  - Variant ID: 3-128904079-G-A
  - RSID: rs115532916
  - Consequence: Missense variant
  - Exome Frequency: 1.82% (AF=0.0182, AC=26,626, AN=1,461,772)
  - Genome Frequency: 1.72% (AF=0.0172, AC=2,615, AN=152,298)
- **Variant Type**: Single nucleotide variant
- **Protein Effect**: p.Ala326Thr (missense substitution)

NM_014049.5(ACAD9):c.976G>A (p.Ala326Thr)

Germline

Classification: **Benign/Likely benign**
7 out of 11 submissions contributed to this classification

	Exomes	Genomes	Total
Filters	Pass	Pass	Discrepant frequencies
Allele Count	26626	2615	29241
Allele Number	1461772	152298	1614070
Allele Frequency	0.01821	0.01717	0.01812
Grpmax Filtering AF (95% confidence)	0.02132	0.02406	0.02152
Number of homozygotes	363	48	411

ClinVar: NM_014049.5(ACAD9):c.976G>A (p.Ala326Thr)

Ref: [link]()

2. ACMG Analysis Agent (BA1 tool calling)

Input

system: You are ACMG Guideline Analyst. You are the 'ACMG Analysis Agent.' You strictly follow the ACMG-AMP 2015 guidelines for genetic variant interpretation. You are responsible for evaluating all 28 criteria—ranging from population data (e.g., BA1, PM2, PS4) to computational predictors, functional assays, segregation, de novo status, and database citations. You apply clinical reasoning using provided tools for each ACMG code and return your output in structured form.
Your personal goal is: Apply all 28 ACMG-AMP 2015 criteria to classify genetic variants using available clinical, population, computational, and functional evidence.
You ONLY have access to the following tools, and should NEVER make up tools that are not listed here:

Tool Name: acmg_ba1_tool
Tool Arguments: {'rsid': {'description': 'Variant rsID', 'type': 'str'}, 'gene': {'description': 'Gene symbol', 'type': 'str'}, 'disease': {'description': 'Associated disease name', 'type': 'str'}, 'gnomad_af': {'description': 'Allele frequency from gnomAD', 'type': 'float'}, 'ba1_threshold': {'description': 'Threshold frequency for BA1 to apply', 'type': 'float'}}
Tool Description: Tool to evaluate BA1 (Benign Standalone) criterion using gnomAD AF and threshold

Output

Thought: I need to evaluate the variant ACAD9 (NM_014049.5):c.976G>A p.(Ala326Thr) using ACMG-AMP 2015 guidelines. First, I'll check the population frequency to see if BA1 or BS1 applies.

Action: acmg_ba1_tool
Action Input: {"rsid": "rs115532916", "gene": "ACAD9", "disease": "ACAD9-related disorder", "gnomad_af": 0.0182, "ba1_threshold": 0.05}

Input

user:
You are an expert ACMG interpretation assistant.

Evaluate if the BA1 criterion (Benign Standalone) should be applied. BA1 applies when a variant is **too common in population databases** (e.g., gnomAD) to be consistent with a pathogenic role.

Variant Info:

- Variant: rs115532916
- Gene: ACAD9
- Disease: ACAD9-related disorder
- Allele Frequency in gnomAD: 0.0182
- BA1 Threshold: 0.05

Instructions:

- If AF > 0.05 in any population (or exceeds disease-specific max AF), BA1 applies.
- If variant is rare or absent, BA1 does not apply.

Respond with:

- BA1: [Yes/No]
- Justification:

Output

- BA1: No
- Justification: The allele frequency (AF) of rs115532916 in gnomAD is 0.0182, which is below the BA1 threshold of 0.05. Since the AF does not exceed the threshold in any population, BA1 does not apply. The variant is considered rare and cannot be dismissed as benign based on population frequency alone.

2. ACMG Analysis Agent (Report generation)

Output

```
Thought: I now know the final answer
Final Answer: {
"criteria_met": {
"BS1": "Yes",
"BP4": "Yes",
"BA1": "No",
"BP6": "No"
},
"classification": "Benign",
"explanation": "The variant rs115532916 in ACAD9 meets BS1 due to its high allele frequency (1.82%) exceeding the disease-specific threshold for ACAD9-related disorder (recessive). BP4 applies as multiple computational tools (SIFT: Tolerated, PolyPhen: Benign) predict a benign effect. BA1 does not apply as the frequency is below the 5% threshold. BP6 does not apply despite ClinVar's Benign/Likely benign classification because public evidence is available. The combined evidence supports a Benign classification."
}
```

Benign -15 points = 2 P - 17 B

- **BP6** Very Strong — Very Strong: ClinVar classifies this variant as Benign, 2 stars (reviewed Mar '25, 11 submissions of which 2 are from high confidence submitters).
- **BS1** Strong — GnomAD exomes allele frequency = 0.0182 is greater than 0.00031 (threshold derived from the 825 clinically reported variants in gene ACAD9), good gnomAD exomes coverage = 46.2.
- **BS2** Strong — Observed in healthy adults: gnomAD genomes homozygous allele count = 48 is greater than 2 for AR gene ACAD9, good gnomAD genomes coverage = 31.9.
- **BP4** Strong — MetaRNN = 0.00957 is between 0.00692 and 0.108 ⇒ strong benign.
- **PM5** Moderate — Alternative variant chr3:128904079 G⇒C (Ala326Pro) is classified Pathogenic, 2 stars, by ClinVar (confirmed using the germline classifier).

3. ACGS Analysis Agent

Input

system: You are ACGS Scoring Specialist.
You are a UK-trained genomics analyst applying the ACGS 2020 Best Practice Guidelines.
You independently analyze a variant's ACMG criteria to compute its point-based classification.

You do not rely on ACMG's final label — instead, you convert the raw criteria into ACGS scores and assign:

- ≥ 10: Pathogenic
- 6–9: Likely Pathogenic
- 0-5: VUS
- -1 to -4: Likely Benign
- ≤ -4: Benign

Use the ACGSScoreTool to transform the list of raw ACMG criteria into a full classification.

Your personal goal is: Apply the ACGS 2020 point-based classification system independently of ACMG result

Output

Thought: I need to calculate the ACGS score based on the provided ACMG criteria (BS1 and BP4) to determine the classification.
Action: calculate_acgs_points
Action Input: {"criteria": ["BS1", "BP4"]}

4. Quality Control Agent

Input

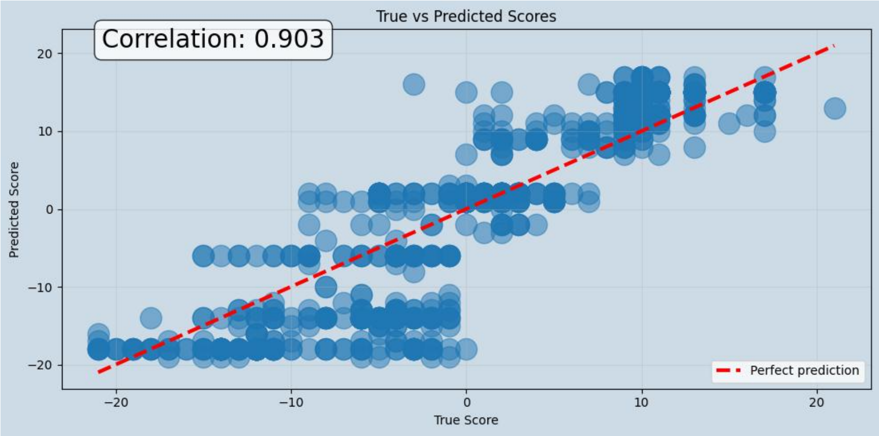
system: You are Quality Control Auditor. You serve as a quality control gatekeeper, ensuring annotations are consistent, justified, and reproducible.
Your personal goal is: Check consistency and conflicts between ACMG and ACGS interpretations and make final call
To give my best complete final answer to the task respond using the exact following format:

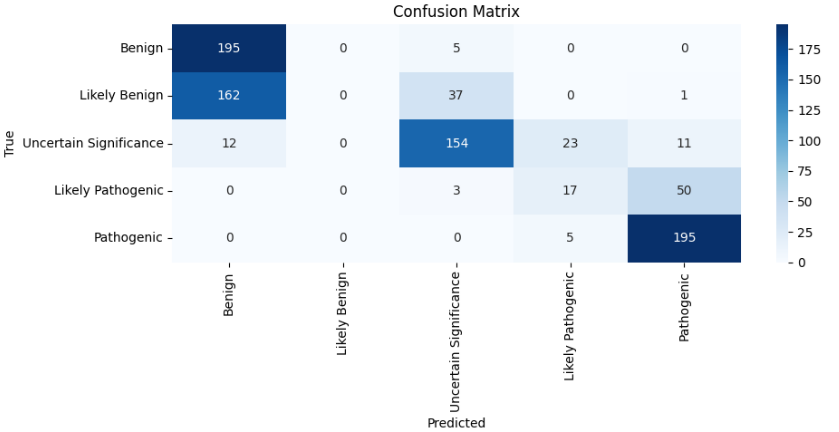
Thought: I now can give a great answer
Final Answer: Your final answer must be the great and the most complete as possible, it must be outcome described.

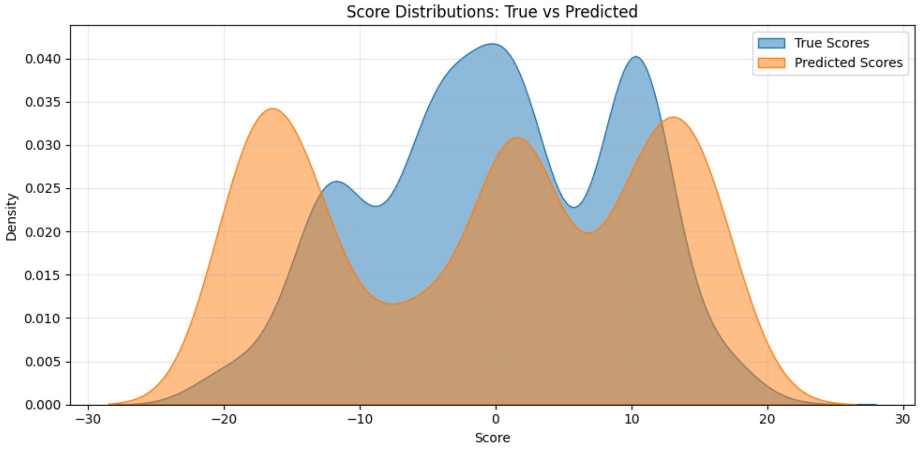
I MUST use these formats, my job depends on it!
user:
Current Task:
Compare ACMG and ACGS interpretations. Highlight any conflicts or misalignments. Provide a final consensus classification and QC justification.

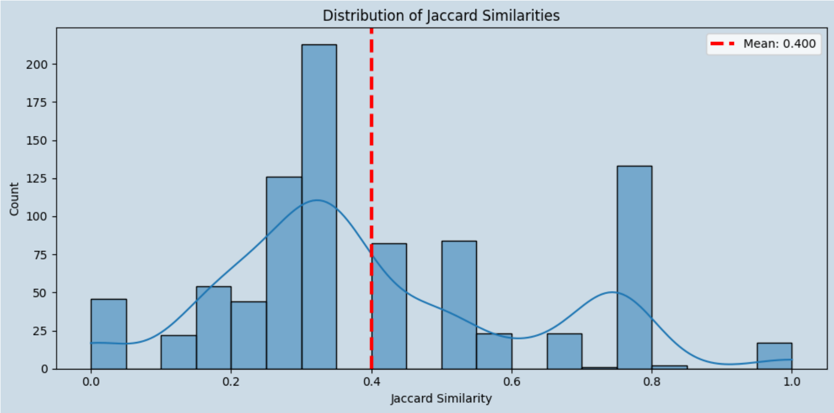

```
"acmg_acgs_comparison": {
  "conflicts": "None",
  "alignment": "Full agreement on Benign classification",
  "acmg_criteria": {
    "BS1": "Yes (allele frequency > disease-specific threshold)",
    "BP4": "Yes (computational evidence supports benign impact)",
    "BA1": "No (frequency <5%)",
    "BP6": "No (public evidence available)"
  },
  "acgs_criteria": {
    "BS1": "Yes (score -4)",
    "BP4": "Yes (score -1)"
  }
},
"final_consensus": {
  "classification": "Benign",
  "qc_notes": "Consistent Benign classification supported by: 1) High population frequency (BS1), 2)
  "justification": "The variant meets Benign criteria through both ACMG (BS1+BP4) and ACGS (total sc
}
```

DeepVAA: Multi-agent with augmented Clinvar, gnomAD data

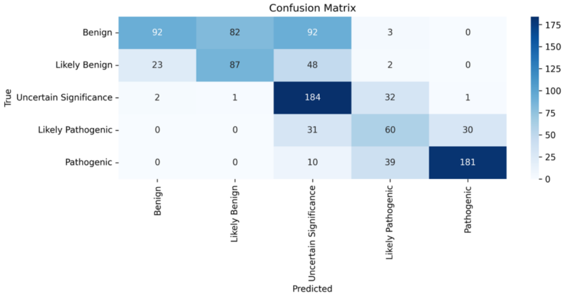
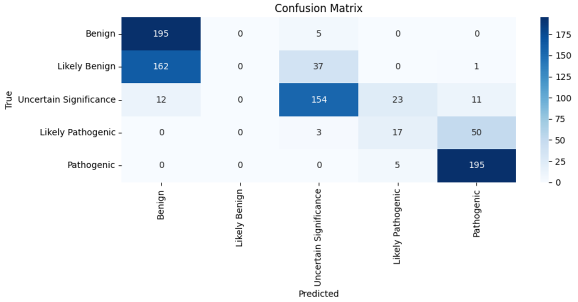








DeepVAA outperformanced Baseline (deepseek-chat)

Model	**Baseline (deepseek-chat)**	**DeepVAA (v1)**
Pearson correlation	0.736	0.903
Jaccard similarity	0.360	0.400
Confusion matrix		

Discussion

- **Future Development**: continue to implement and refine specific ACMG criteria checks within the system.
- **Agent Refinement**: improve the instructions and reasoning capabilities of the individual agents.
- The DeepVAA project aims to enhance the efficiency and accuracy of genetic variant interpretation, thereby streamlining clinical genomics workflows and improving the reliability of variant classification in medical practice.

Current work: 28 criteria of ACMG guidelines

Solved

Criterion	Classification_1	Strength	ACGS_Points
PM2	Pathogenic	Moderate	2
PP3	Pathogenic	Supporting	1
BP4	Benign	Supporting	-1
BP7	Benign	Supporting	-1
BP6	Benign	Supporting	-1
BS2	Benign	Strong	-4
BS1	Benign	Strong	-4
BA1	Benign	Stand-alone	-8

In-progress

Criterion	Classification_1	Strength	ACGS_Points
PVS1	Pathogenic	Very Strong	8
PS1	Pathogenic	Strong	4
PS3	Pathogenic	Strong	4
PM1	Pathogenic	Moderate	2
PM5	Pathogenic	Moderate	2
PP5	Pathogenic	Supporting	1
PP2	Pathogenic	Supporting	1
BP1	Benign	Supporting	-1
BP3	Benign	Supporting	-1
BS3	Benign	Strong	-4

Future-works

Criterion	Classification_1	Strength	ACGS_Points
PS2	Pathogenic	Strong	4
PS4	Pathogenic	Strong	4
PM3	Pathogenic	Moderate	2
PM4	Pathogenic	Moderate	2
PM6	Pathogenic	Moderate	2
PP1	Pathogenic	Supporting	1
PP4	Pathogenic	Supporting	1
BP2	Benign	Supporting	-1
BP5	Benign	Supporting	-1
BS4	Benign	Strong	-4

Current work: 28 criteria of ACMG guidelines

	Benign: Strong	Benign: Supporting	Pathogenic: Supporting	Pathogenic: Moderate	Pathogenic: Strong	Pathogenic: Very strong
Population data	MAF is too high for disorder BA1/BS1 **OR** observation in controls inconsistent with disease penetrance BS2			Absent in population databases PM2	Prevalence in affecteds statistically increased over controls PS4	
Computational and predictive data		Multiple lines of computational evidence suggest no impact on gene /gene product BP4 Missense in gene where only truncating cause disease BP1 Silent variant with non predicted splice impact BP7 In-frame indels in repeat w/out known function BP3	Multiple lines of computational evidence support a deleterious effect on the gene /gene product PP3	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before PM5 Protein length changing variant PM4	Same amino acid change as an established pathogenic variant PS1	Predicted null variant in a gene where LOF is a known mechanism of disease PVS1
Functional data	Well-established functional studies show no deleterious effect BS3		Missense in gene with low rate of benign missense variants and path. missenses common PP2	Mutational hot spot or well-studied functional domain without benign variation PM1	Well-established functional studies show a deleterious effect PS3	
Segregation data	Nonsegregation with disease BS4		Cosegregation with disease in multiple affected family members PP1	Increased segregation data →		
De novo data				De novo (without paternity & maternity confirmed) PM6	De novo (paternity and maternity confirmed) PS2	
Allelic data		Observed in *trans* with a dominant variant BP2 Observed in *cis* with a pathogenic variant BP2		For recessive disorders, detected in trans with a pathogenic variant PM3		
Other database		Reputable source w/out shared data = benign BP6	Reputable source = pathogenic PP5			
Other data		Found in case with an alternate cause BP5	Patient's phenotype or FH highly specific for gene PP4			

Solved

In-progress

References

[1] "A review of genetic variant databases and machine learning tools for predicting the pathogenicity of breast cancer," 2025.

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[3] "LEAP: Using machine learning to support variant classification in a clinical setting," 2025.

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[6] "A comprehensive review of deep learning-based variant calling methods," 2025.

[7] "Diagnostic Classification and Prognostic Prediction Using Common Genetic Variants in Autism Spectrum Disorder" 2025.

[8] "The future role of facial image analysis in ACMG classification guidelines," 2025.

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[11] "ACMG-AMP updated guidelines and engaging underrepresented groups in ClinGen main session," 2025.

[12] "ACMG Standards Guidelines for the Interpretation of Sequence," 2025.

Thank you for your attention!
Q&A

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